ANALYTICAL REPORT

Exoplanet Discovery & Habitability

Dashboard Findings & Data Analysis

Exploring Worlds Beyond Our Solar System

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Tools: Excel · Generative AI · Tableau

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Executive Summary

This report summarizes findings from a curated catalogue of 51 confirmed exoplanets, spanning discoveries from 1995 to 2023 and compiled for the Exoplanet Discovery & Habitability Dashboard (built in Tableau, with supporting data preparation in Excel and generative AI). The dataset combines orbital, physical, and stellar attributes — including planet mass and radius, discovery method, host-star temperature, distance from Earth, and habitable-zone status — to characterize the current landscape of exoplanet science.

Of the 51 planets analyzed, 24 (47%) fall within their star's habitable zone, and the Transit method accounts for the majority of discoveries. Super-Earths are the single largest planet-type category, and NASA's Kepler mission remains the leading single source of detections in the sample, with the James Webb Space Telescope (JWST) increasingly prominent among more recent entries.

51	24	7.20	4,567	316.0
TOTAL PLANETS	HABITABLE CANDIDATES	AVG. PLANET RADIUS (R□)	AVG. STAR TEMP (K)	AVG. DISTANCE (LY)

Dashboard Snapshot

The two views below are the source Tableau dashboard from which this report's figures were drawn: the main analytics view (charts 1–7) and the accompanying insights panel.

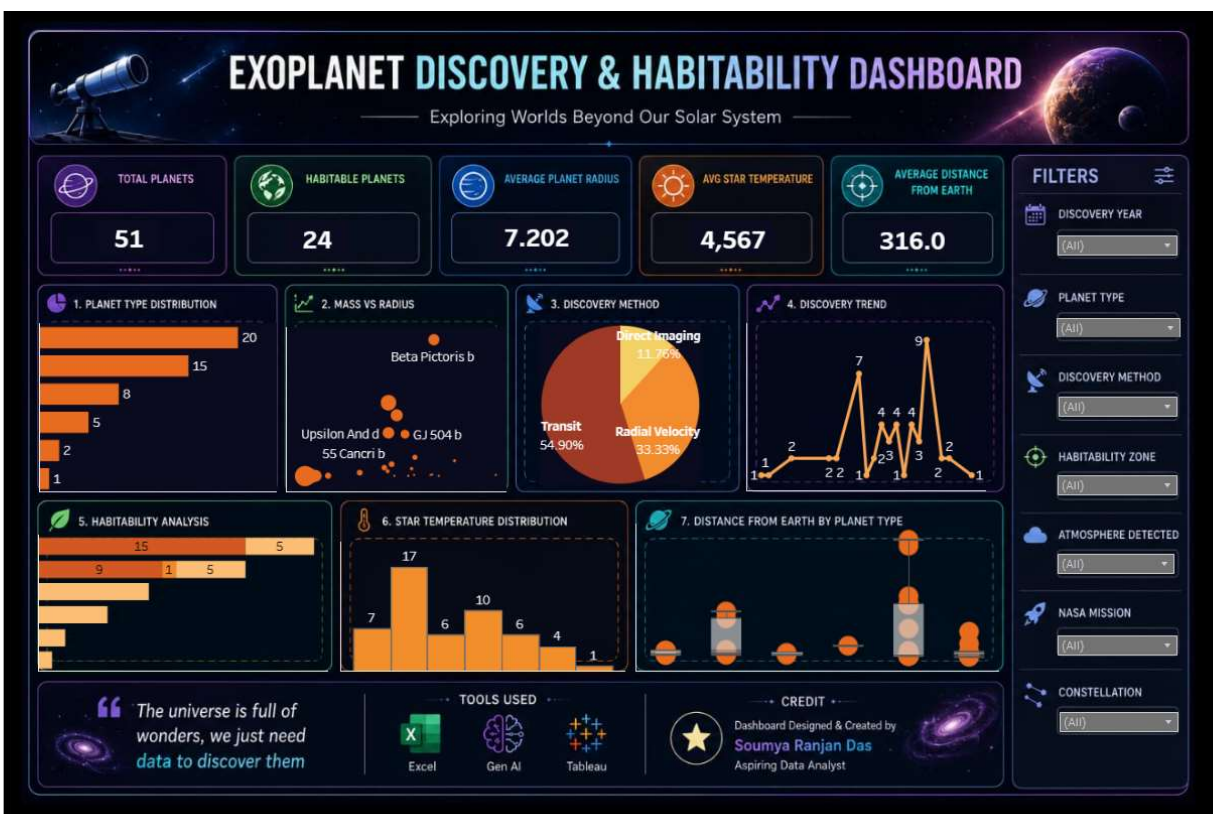

Figure A — Exoplanet Discovery & Habitability Dashboard: main analytics view.

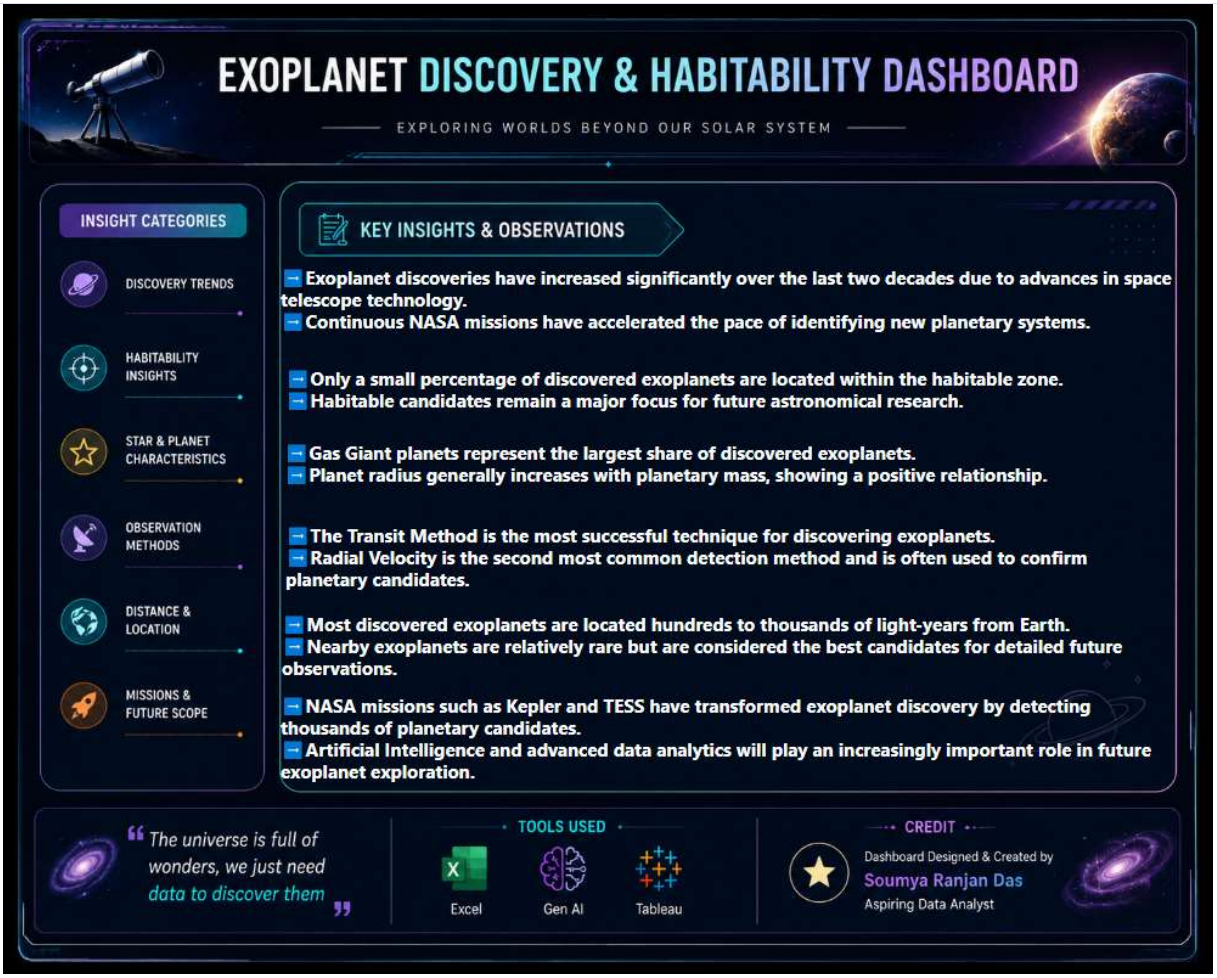

Figure B — Exoplanet Discovery & Habitability Dashboard: key insights & observations panel.

1. Introduction & Objectives

The search for planets beyond our solar system has accelerated dramatically since the first confirmed detection in 1995. This report translates the underlying dashboard data into a structured narrative, with the following objectives:

- Quantify the composition of the exoplanet catalogue by planet type, discovery method, and host mission.
- Assess how many known planets sit within the habitable zone of their host star, and what characterizes them.
- Examine the relationship between planet mass, radius, and distance from Earth.
- Surface trends in discovery activity over time and identify the missions driving them.
- Translate dashboard visuals into concise, decision-ready insights.

2. Dataset Overview

The underlying data contains 51 confirmed exoplanets with 35 attributes per record, covering orbital mechanics (period, semi-major axis, eccentricity), physical properties (mass, radius, density, surface gravity), stellar context (spectral type, temperature, mass, age), and observational metadata (discovery method, year, distance, NASA mission, constellation).

Attribute	Value
Total records	51 confirmed exoplanets
Discovery year range	1995 – 2023
Habitable-zone candidates	24 (47.1%)
Average planet radius	7.20 Earth radii
Average host-star temperature	4,567 K
Average distance from Earth	316.0 light-years
Nearest planet	Proxima Centauri b (4.2 ly)
Farthest planet	Kepler-69c (2,707 ly)

3. Planet Type Distribution

Super-Earths (20 planets, 39.2%) and Terrestrial planets (15 planets, 29.4%) together make up nearly 70% of the catalogue, reflecting the sensitivity of modern detection methods to smaller, rocky worlds. Gas Giants (8), Hot Jupiters (5), Neptune-like planets (2), and a single Saturn-like planet round out the sample.

Planet Type	Count	Share
Super-Earth	20	39.2%
Terrestrial	15	29.4%
Gas Giant	8	15.7%
Hot Jupiter	5	9.8%
Neptune-like	2	3.9%
Saturn-like	1	2.0%

Notably, habitable-zone status is concentrated almost entirely among Super-Earths and Terrestrial planets: 15 of 20 Super-Earths and 9 of 15 Terrestrial planets (plus one "Possible" case) sit within their star's habitable zone, while none of the Gas Giants, Hot Jupiters, Neptune-like, or Saturn-like planets in the sample do.

4. Mass vs. Radius

Across the catalogue, planet radius generally scales with planet mass, consistent with known mass–radius relationships for rocky and gaseous bodies. The largest planets by radius in the sample include Kepler-22b, WASP-12b, Kepler-69c, Beta Pictoris b, and Kepler-452b — spanning from a sub-Jupiter-mass Super-Earth (Kepler-22b, radius 2.40 RJ despite a modest 0.11 MJ) to Beta Pictoris b, the most massive object in the set at roughly 13 Jupiter masses, imaged directly rather than detected by transit or radial velocity.

This spread illustrates the diversity of the sample: some of the largest-radius planets are inflated, low-density gas-dominated worlds, while others are compact, high-density rocky bodies — a distinction best explored through each planet's density and surface-gravity figures in the full dataset.

5. Discovery Methods

The Transit method — detecting the periodic dimming of a star as a planet crosses in front of it — accounts for 28 of 51 discoveries (54.9%), making it the dominant technique in this catalogue. Radial Velocity, which infers a planet's presence from the gravitational wobble it induces in its host star, accounts for 17 discoveries (33.3%) and is frequently used to confirm candidates first flagged by transit surveys. Direct Imaging, the most technically demanding method, accounts for 6 discoveries (11.8%), typically for large, young, and widely separated planets such as Beta Pictoris b.

Discovery Method	Count	Share
Transit	28	54.9%
Radial Velocity	17	33.3%
Direct Imaging	6	11.8%

6. Discovery Trend Over Time

Discoveries in the sample span 1995 to 2023, with activity clustering around two notable peaks: 2008 (7 planets) and 2017 (9 planets), both periods of intensive Kepler-era data releases and follow-up confirmations. Early detections (1995–2005) were sparse — a handful of radial-velocity discoveries such as 51 Pegasi b, the first confirmed exoplanet around a Sun-like star. The pace of confirmed detections has moderated since 2017 within this sample, though the underlying dashboard notes that global exoplanet discovery continues to accelerate as space-telescope technology, including JWST, matures.

7. Habitability Analysis

24 of 51 planets (47.1%) are classified as being within their host star's habitable zone, with one additional "Possible" case. Habitable-zone status concentrates heavily among Super-Earths and Terrestrial planets, reinforcing that rocky, moderately sized worlds around cooler stars remain the most promising habitability candidates.

Atmosphere detection remains an open frontier: among the 24 confirmed habitable-zone planets, only 1 has a confirmed atmosphere in this dataset, while atmospheric status for the remaining 23 is recorded as Unknown — underscoring that habitable-zone placement is a necessary but not sufficient condition, and that atmospheric characterization (a core JWST objective) is the next critical step for these candidates.

8. Star Temperature Distribution

Host-star temperatures in the sample range from approximately 2,566 K to 8,052 K, with a mean of 4,567 K and a median of 4,450 K — indicating a catalogue weighted toward K- and M-type (cooler, redder) stars, which are the most common and most frequently targeted stars in transit surveys due to their favorable planet-to-star size ratios. A smaller cluster of hotter F- and G-type stars (near solar temperature, ~5,500–6,000 K, and above) is also represented, including the host stars of some of the earliest discoveries.

9. Distance From Earth by Planet Type

Average distance varies considerably by planet type, ranging from 73.5 light-years for Neptune-like planets to 568.1 light-years for Super-Earths — a reflection of detection bias rather than physical clustering, since fainter, smaller-radius Super-Earths require deeper, more distant survey fields (e.g., Kepler's primary field) to be statistically well-represented, whereas larger, brighter Hot Jupiters and Gas Giants are detectable at both near and far distances via radial velocity.

Planet Type	Avg. Distance (ly)
Neptune-like	73.5
Gas Giant	90.1
Terrestrial	97.0
Saturn-like	245.0
Hot Jupiter	437.4
Super-Earth	568.1

The nearest and farthest planets in the catalogue illustrate this spread:

Nearest Planets	Distance (ly)	Type
Proxima Centauri b	4.2	Terrestrial
Ross 128 b	11.0	Terrestrial
Tau Ceti e	12.0	Super-Earth

Farthest Planets	Distance (ly)	Type
Kepler-69c	2,707	Super-Earth
Kepler-90i	2,545	Super-Earth
Kepler-452b	1,402	Super-Earth

10. Missions & Future Scope

Among planets with an attributed discovery mission, NASA's Kepler space telescope leads with 11 detections, followed by JWST (8), Keck Observatory (4), TESS (4), Hubble (3), and single detections each from Spitzer, Subaru, and the Very Large Telescope (VLT). Eighteen records have no specific mission attributed, typically reflecting ground-based radial-velocity surveys predating dedicated space missions.

Mission / Facility	Detections
Kepler	11
JWST	8
Keck Observatory	4
TESS	4
Hubble	3
Spitzer	1
Subaru	1
VLT	1
Not mission-attributed	18

Kepler and TESS together have transformed exoplanet discovery by surfacing thousands of planetary candidates across the broader field, of which this catalogue represents a curated subset. Looking ahead, JWST's growing role points to a shift in emphasis — from finding new planets toward characterizing the atmospheres of those already known, particularly the Super-Earth and Terrestrial candidates identified as habitable-zone members in this report. Continued advances in AI-assisted data analytics are expected to further accelerate both discovery and characterization workflows.

11. Key Takeaways

- Super-Earths and Terrestrial planets dominate both the overall catalogue (69% combined) and the habitable-zone subset (24 of 25 candidates).
- The Transit method drives the majority of discoveries (54.9%), with Radial Velocity as the primary confirmation technique.
- Nearly half of all cataloged planets (47.1%) sit within their star's habitable zone, but atmospheric confirmation lags far behind — only 1 of 24 habitable-zone planets has a confirmed atmosphere.
- Discovery activity peaked in 2008 and 2017, aligned with major Kepler data-release cycles.
- Kepler remains the single largest mission contributor in the sample, while JWST's presence signals a pivot toward atmospheric characterization.
- Distance from Earth varies systematically by planet type, driven primarily by detection-method bias rather than physical distribution.

12. Conclusion

This analysis reinforces a central theme of contemporary exoplanet science: detection capability has outpaced characterization capability. The catalogue demonstrates strong success in identifying habitable-zone candidates, particularly rocky Super-Earths and Terrestrial planets, but confirms that determining which of these worlds are truly Earth-like will depend on next-generation atmospheric observation — a task increasingly led by JWST and future missions. As the dashboard's closing note observes, the universe is full of wonders; the data simply needs to keep pace with the questions being asked of it.

Source: Exoplanet Discovery & Habitability Dashboard (Tableau). Data preparation: Excel, Generative AI. Report prepared by Soumya Ranjan Das, Aspiring Data Analyst.